

Wireless Communication (EX751 / EX715)

Concise PYQ

Notes: TU IOE PYQ 2070–2082, 22 papers. Regular = **bold**, Back = normal; BEX (EX751) = normal font, BEI (EX715) = **mono**. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Evolution of wireless / cellular 1G–5G; technology + worldwide market penetration (4)(6)
- 1G/2G/3G/4G Vs, technology advancement; features 4G over 3G; benefits of wireless (4)(5)(6)(2+4)
- Trends beyond 3G; forward & reverse channel; 5G Vs 4G, features of 5G (4)(2)(2+2)

2. Cellular Concepts

- Frequency reuse; co-channel reuse ratio $Q=\sqrt{3N}$ (prove); find N (omni/120°/60°) (6)(4)(3+3)(4+4)(3+5)
 - $S/I=\sqrt[3]{3N}/i_0$; min. co-channel distance (7-cell, 23 km²); $N=7$ concept (4)(3)
- Handoff — margin +Fig, GSM strategy, types & application, practical considerations (3)(4)(6)(8)
 - Proper Vs improper handoff +Fig; cell dragging; prioritized handoff schemes (1+3)(2+4)(3+4)
- Co-channel Vs adjacent interference; cell footprint, interference tier (2.5)(1+4)(4)
- GoS + trunking (blocked-call-cleared); traffic intensity, Erlang-B tables (3+5)(2+2+2+3)
 - 394 cells/19 ch/2% blocking → users + market penetration (5)
 - 1500 km², 12 cells, 28.5 MHz → cells/channels/traffic/users (12)
 - 32 cells, 1.6 km, reuse 7, 336 ch → area, ch/cell, simultaneous calls (4)
 - $N=7$, 1% blocking, 57 ch, omni → 60° sectorized: capacity loss (2+2+2)
 - cluster size / clusters / max. users — *73 Ma prints no data* (6)
- Microcell zone concept; sectoring Vs cell splitting; capacity & coverage enhancement (3)(5)(8)

3. Radio Wave Propagation

- Free-space & link budget — received power / E-field / flux density / rms voltage (8)(6)
 - 50 W@900 MHz, 10 km, $G_T=1$, $G_R=2$, 50 Ω; 165 W@325 MHz, 15 km
 - 10 W@250 MHz + 20 m coax + mismatch; 150 W, -104 dBm threshold, $n=4$
 - 50 W → dBm/dBW, $P_r(100\text{ m})$, $P_r(10\text{ km})$; cordless 52000 kHz, 50 m
 - $\lambda/4$ monopole 2.55 dB, $E=10^{-3}$ V/m@1 km, 2-ray; far-field, 1 m@900 MHz
- Propagation mechanisms (3 basic); large- Vs small-scale; path loss + multipath params (3)(3+6)(8)(2+4)
- Two-ray path/phase difference; Fresnel diffraction; scattering + ground reflection (6)(8)(2+6)(2+8)
- Log-distance model; link budget design (BTS↔mobile reverse link; 10-km AP link) (4)(10)(12)
- Outdoor models — Okumura Vs Hata, necessary conditions (2+2)(3+3)(5)
 - Okumura: $d=50\text{ km}$, $h_{te}=100\text{ m}$, $h_{re}=10\text{ m}$, EIRP 1 kW@900 MHz, $A_{mu}=43$, $G_{area}=9$ (4+2+2)
 - Okumura: find T–R separation, $L=167\text{ dB@2.1 GHz}$, $A_{mu}=34$, $G_{area}=0$ (8)
 - Hata: medium city 950 MHz/45 m/10 km/5 m; large urban 900 MHz/100 m/2 m/4 km (4+4)(7)
 - Hata: max. cell radius & area, 50 W, -91 dBm, 900 MHz, 30 m/1 m (6)
 - Two-ray point-to-point 800 MHz/30 m/2 m/10 km; Ericsson Multiple Breakpoint (4+4)(2)
- Indoor models (any two) + influencing factors (5)(8)
- Small-scale fading types + influencing factors; Doppler shift derivation (2+2)(4+4)(2+4+4)
- Doppler spread; coherence BW & time (I); rms delay spread; fading classification (4)(2+5)(2+7)(4+4)
 - PDP numericals: T_s without equalizer ($\sigma_\tau/T_s \leq 0.1$); 90%/50% coherence BW

- $P_r(\tau)=(-20, -10, -10, 0)\text{ dB @ } \tau=(0, 1, 2, 5)$; $(0, -10, -20, -23)\text{ dB @ } (0, 100, 200, 400)\text{ ns}$
- flat Vs frequency-selective for AMPS (30 kHz) & GSM (200 kHz) (6)(8)
- Rayleigh Vs Rician fading distribution; frequency selective fading (2)(5)(2+4)

4. Modulation–Demodulation

- QPSK / OQPSK / $\pi/4$ -QPSK — Tx & detection +constellation; BPSK Vs QPSK (6)(7)(8)(5+2)
- DPSK Tx/Rx + PN sequence (I) (4+2+2)
- MSK Vs GMSK Vs OQPSK (BW/power efficiency, GSM); GMSK Tx & detection (3+3+2)(2+6)(8)
- OFDM operation +Fig (Tx and Rx separately); (+)(-); M-ary QAM, 16-QAM Vs 16-PSK (4)(8)(4+4)(2+6)
- Spread spectrum — define, DSSS +Fig, FHSS +Fig; (+) and major (-) (2+5)(2+6)(4)

5. Equalization & Diversity

- Equalization (I), signal processing to minimize ISI; linear transversal equalizer (4)(2+5)(2+1+4)
- Training & tracking modes; adaptive equalization algorithms; MSE optimal weights (1+3)(4)(7)(8)
- MLSE +Fig; interleaving — need, working principle +Fig (4)(6+1+5)(2+6)
- Diversity (I) + types; combining (max-ratio / feedback / space); antenna diversity Vs (2+2)(2+6)(4+6)(3+3+2)
- RAKE (RACK) receiver, M-branch +Fig; time diversity (3)(4)(8)(3+5)

6. Speech & Channel Coding

- Speech signal characteristics; used in designing coders (3)(4)(6)(8)
- Vocoders +Fig, kinds; formant vocoder; frequency-domain coding; GSM CODEC (2+3)(4+4)(3+5)
- LPC +block diagram, types; sub-band coding (Tx/Rx); source coders (2+6)(3+5)(6)
- Channel coding; 7-bit Hamming +Eg; convolutional encoding/decoding (2)(5)
- Viterbi decoding; turbo coding (2)(3)(5)

7. Multiple Access

- FDMA — non-linear effect; no. of channels (25 MHz/100 kHz/200 kHz; 12.8 MHz & 12.5 MHz/10 kHz/30 kHz); (+)(-) (3+2)(4)
- TDMA — (+) over FDMA; GSM frame efficiency; N-TDMA users/cluster (25 MHz/30 kHz/ B_g 20 kHz) (4)
- CDMA — features/(+)(-); design considerations; encoding & decoding +Eg (2+6)(4)(7)(8)
- Hadamard H_8 / 8×8 Hadamard code (Walsh), conditions (3)(4)(3+5)
- SSMA (DS & FH) variants; FHMA +Fig (Tx/Rx) +Eg, types (4+4)(6)(7)(8)
- Hybrid SS MA (any two, near-far mitigation); self-jamming; near-far effect + solutions (2+2)(3+4)(5+4)(6)(2+4+6)
- SDMA; multiple access Vs (FDMA/TDMA/CDMA/SDMA, TCDMA, TDFH) (2+6)(4)(7)
- WLAN; WiFi (3)(5)

8. Wireless Systems & Standards

- GSM architecture (each component, NSS); traffic & control channels; BCH & CCCH (4)(5)(8)(4+4)
- GSM frame structure & hierarchy; TDMA frame efficiency < 100%; channel structure (4+4)(4+6)(6+2)
- GSM signal processing (speech ↔ radio); GSM specifications (5)(8)
- CDMA IS-95 forward & reverse channels (draw); pilot & sync channels; channelization code (4)(6)
- GSM Vs CDMA standards; CDMA Vs LTE architecture (3)(3+5)
- WiMAX; LTE; duplexing + types (3)(4)(5)(2+4)
- Regulatory issues — spectrum allocation, licensing, spectrum management (4)(5)(2+3)(2+4)